

Linux File Permissions and User Access

1. Create user:

```
sudo adduser moniuser
```

2. Change ownership:

```
sudo chown -R moniuser:moniuser /opt/container-monitor
```

3. Verify:

```
ls -ld /opt/container-monitor
```

4. Set permission:

```
sudo chmod -R 700 /opt/container-monitor
```

5. Verify again:

```
ls -ld /opt/container-monitor
```

6. Create another user:

```
sudo adduser testuser
```

7. Switch:

```
su testuser
```

8. Test access:

```
cd /opt/container-monitor
```

Expected:

Permission denied

```
ubuntu@ip-172-31-34-169:~$ ls
docker-project
ubuntu@ip-172-31-34-169:~$ \sudo adduser moniuser
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for moniuser
Enter the new value, or press ENTER for the default
  Full Name []: Naveen
   Room Number []: 100
    Work Phone []:
   Home Phone []:
        Other []:
Is the information correct? [Y/n] y
```

```
ubuntu@ip-172-31-34-169:~$ sudo chown -R moniuser:moniuser /opt/container-monitor
ubuntu@ip-172-31-34-169:~$ ls -ld /opt/container-monitor
drwxr-xr-x 3 moniuser moniuser 4096 Jul 11 10:07 /opt/container-monitor
ubuntu@ip-172-31-34-169:~$ sudo chmod -R 700 /opt/container-monitor
ubuntu@ip-172-31-34-169:~$ ls -ld /opt/container-monitor
drwx----- 3 moniuser moniuser 4096 Jul 11 10:07 /opt/container-monitor
ubuntu@ip-172-31-34-169:~$ sudo adduser testuser
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for testuser
Enter the new value, or press ENTER for the default
  Full Name []:
   Room Number []:
    Work Phone []:
   Home Phone []:
        Other []:
Is the information correct? [Y/n] y
ubuntu@ip-172-31-34-169:~$ su testuser
Password:
testuser@ip-172-31-34-169:/home/ubuntu$ cd /opt/container-monitor
bash: cd: /opt/container-monitor: Permission denied
testuser@ip-172-31-34-169:/home/ubuntu$
```